

Lab 4: VibeCoding - Chat History

Project: 07_catch_falling (Pygame Catch the Falling Objects)

Tool: Claude (Anthropic)

Features built through prompting: collision fix (height check, no remove-while-iterating), fully on-screen smooth basket, varied capped spawning, SPACE speed boost with cooldown and on-screen indicator.

Part 1 - Setup and before video

USER:

I gave Claude the lab handout and the four README tasks (fix collision bug, basket control, controlled spawning, temporary speed boost) and asked it to act as a pair-programming partner: one step at a time, short replies, complete files only, 2 to 4 attempts in total, exact Windows cmd and Notepad instructions, then the Lab-4 git push and a chat_history.pdf at the end. Setup: Windows, Python 3.13, pygame 2.6.1, repo ChiragBasanth/07_catch_falling (no push or PR to SETAPESU26).

AI (Claude):

Step 1 gave the clone, cd, pip install and python main.py commands, the git config commands, and the Win+Alt+R instructions for recording the 10-second before video showing the bugs (basket half off the edge, objects scoring instantly at the top).

USER:

Asked for only the git config statements, then supplied the git email. Recorded the before video and continued.

AI (Claude):

Gave the two git config --global commands (user.name ChiragBasanth and the GitHub email), then a one-line command that copies all source files to the clipboard for pasting into the chat.

Part 2 - Prompt 1 (all four tasks in one go)

USER:

Pasted the full original code (basket.py, collision.py, falling_object.py, game_engine.py, renderer.py, main.py) and asked for all task updates at once, not one task at a time, so each file could be replaced in Notepad and lead straight to the final version.

AI (Claude):

Four files changed: collision.py, basket.py, game_engine.py and renderer.py. main.py, falling_object.py and __init__.py stayed as they were.

- collision.py (Task 1): a catch now requires the object to reach basket height, so nothing scores at the top.
- game_engine.py (Tasks 1, 3, 4): the catch loop now rebuilds the list instead of removing while iterating, so every object is credited. Spawn interval is random (25 to 70 frames), x positions are varied inside the play area and avoid the last spawn spot and nearby objects, and at most 6 objects are on screen. SPACE triggers the boost.
- basket.py (Tasks 2, 4): move() clamps using the basket's half-width so it stays fully on screen and uses smoothed velocity for smooth movement. Boost doubles speed for 3 seconds, then a 5-second cooldown, and speed returns to normal automatically.

- `renderer.py` (Task 4): the basket turns cyan while boosted and a status bar shows BOOST ACTIVE, Boost recharging or Boost READY [SPACE].

[Files delivered: `collision.py`, `basket.py`, `game_engine.py`, `renderer.py`]

A verification checklist was given: basket fully on screen at both edges, no instant scoring, varied spawns capped at 6, boost indicator and expiry, Game Over after 5 misses and R to restart.

Testing after Prompt 1

The game ran and the checks were tried. One bug was found: while the boost was on, objects that visually touched the basket were not being caught. This led to Prompt 2.

Part 3 - Prompt 2 (boost catch bug)

USER:

when boost is on its not capturing even if touching why

AI (Claude):

Two causes were found. First, the catch check tested only the object's center against the basket's side edges, so a ball touching the side did not count. Second, the white boost outline was drawn 4 pixels larger than the real hitbox, so objects looked like they touched when they did not. Two files changed:

- `collision.py`: replaced with a circle-versus-rectangle test using the nearest point on the basket and the object's radius.
- `renderer.py`: the boost outline is now drawn inside the basket hitbox so visuals match collisions.

[Files delivered: `collision.py`, `renderer.py`]

Testing after Prompt 2

The game was re-run with boost active, catching with the basket's edges and middle. The after video (PES1UG24CS133_after.mp4) was recorded showing the cyan boosted basket, the boost status bar, varied spawn positions and timing, the basket staying fully on screen, and objects no longer scoring at the top. The before video is PES1UG24CS133_before.mp4.

Part 4 - Submission steps

The final steps were: copy the code folder and both videos into a Lab-4 folder inside the local clone, export this chat as `chat_history.pdf` into the same folder, then `git add Lab-4`, `git commit` and `git push origin main` to ChiragBasant/07_catch_falling (with `git pull --rebase origin main` first if the push is rejected with fetch first), without raising a PR to SETAPESU26.